Rachakonda Abhinay

AI/ML Engineer

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OBJECTIVE

Generative AI Engineer with hands-on experience in building RAG pipelines, LLM-based applications, and deploying scalable AI systems using FastAPI and AWS. Passionate about solving real-world problems using AI

Education

OSMANIA UNIVERSITY,MBA 2023 – 2025 | Hyderabad, Telangana
MANTRA BUSSINESS SCHOOL OF MANAGEMENT
CGPA: 86%

Skills

Programming & Libraries — Python,BeautifulSoup, NumPy, Pandas, Matplotlib, Seaborn, NLTK, Scikit-learn, OpenCV, TensorFlow, PyTorch.

Machine Learning — Supervised & Unsupervised Learning, Feature Engineering, Data Preprocessing, Hyperparameter Tuning, Model Evaluation.

Deep Learning — ANN, CNN, RNN,LSTM,GRU

Generative AI and Agent Frameworks — Transformers architecture ,LangChain, LangGraph, CrewAI, Agno, AutoGen, DeepAgents,Hugging Face Fine-Tuning, Prompt Engineering,Agentic_RAG

Deployment & MLOps: — FastAPI, Docker, MLflow,AWS EC2, Model & Application Deployment

EXPERIENCE

Geeth LLC — AI/ML Engineer Feb 2025 – Feb 2026
Remote,Waukesha, Wisconsin,Usa

- Architected an **Agentic RAG pipeline** for multi-format document QA using FAISS and LLMs, improving retrieval efficiency by ~90%.
- Implemented an **AI-powered ATS Resume Screening System** for resume ingestion and JD-based scoring, with LLM-generated interview questions, reducing manual effort by ~95%.
- Engineered a **Voice Assistant system** integrating speech-to-text → LLM → text-to-speech pipeline for real-time interaction.
- Created a **Video Summarization pipeline** using Whisper transcription and LLM-based summarization for long-form content.
- Integrated **Plaid APIs via FastAPI microservices** to securely retrieve financial data (transactions, balances, identity), deployed using Docker.
- Designed an **AI Decision Intelligence Platform** for automated data profiling, visualization, and insight generation.
- Constructed an **ETL pipeline** for financial data extraction and reconciliation using OCR and rule-based matching.
- Orchestrated a **multi-provider LLM platform** with MongoDB-based conversation logging and prompt optimization.
- Established a **LoRA-based fine-tuning pipeline** for code generation models using Transformers and PyTorch.

Innomatics Research Labs - Data Science Intern Jan 2024 – Mar 2024
Hyderabad, India

- Built an AI-powered **Code Reviewer** using Streamlit and OpenAI APIs for automated code analysis.
- Developed a **Conversational Data Science Tutor** using Gemini for interactive ML explanations.
- Implemented **RAG pipelines** using LangChain to improve knowledge retrieval.
- Optimized a **video subtitle search engine** using NLP techniques.
- Set up **ML pipelines** with MLflow and Prefect for experiment tracking and model management.
- Deployed a **sentiment analysis system** using Flask on AWS EC2.

Projects

Multimodal RAG System

- Built an advanced agentic multimodal system capable of indexing and querying diverse data sources, including structured and unstructured formats.
- Implemented hybrid retrieval by combining semantic search (vector-based) with keyword-based search to significantly improve retrieval accuracy and relevance.
- Utilized MongoDB for efficient storage and management of metadata, document chunks, and indexing references, enabling fast and scalable data access.
- Integrated Groq LLM for high-speed, context-aware response generation, ensuring low latency and high-quality outputs.
- Applied RAG evaluation techniques inspired by RAGAS to measure key performance.

Technologies: Python, LangChain, FAISS, MongoDB, Groq LLM, FastAPI, OCR

Movie Recommendation System using Machine Learning

- Developed a content-based movie recommendation engine using the TMDB dataset.
- Performed NLP preprocessing including tokenization, stopword removal, stemming, and lemmatization using NLTK.
- Extracted textual features using CountVectorizer and computed similarity using Cosine Similarity.
- Built and deployed an interactive Streamlit web application on AWS EC2.
- **Technologies:** Python, Pandas, NLTK, Scikit-learn, CountVectorizer, Cosine Similarity, Streamlit, AWS EC2

Lung Cancer Detection Using VGG16

- Developed a deep learning model to classify lung cancer types including lung squamous cell carcinoma, lung adenocarcinoma, and normal lung tissue using histopathological images.
- Applied image preprocessing and augmentation using ImageDataGenerator. Leveraged a pretrained VGG16 model with frozen base layers and added custom dense layers for classification.
- Trained the model using categorical cross-entropy loss with the Adam optimizer and evaluated performance using confusion matrix and classification reports.
- **Technologies:** Python, TensorFlow, Keras, OpenCV, Matplotlib, Seaborn

Courses

Data Science - innomatics research labs Jun 2023 – Jun 2024 | Hyderabad, India

Certificates

- Machine Learning using python Libraries of IBM
- Advance Machine Learning and Generative Ai internship program